



Unit 4 · Outcome 2 · Energy Sources

The seven renewables, and why they never run out

KK02

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



What makes a source "renewable"?

The whole study design splits energy into non-renewable (KK01) and renewable (KK02). The line between them is not "does it pollute" or "is it clean" — it is about rate.

A renewable energy source is replenished by natural processes at a rate equal to or faster than the rate at which it is used, so it is not depleted on a human timescale.

Non-renewable sources (coal, oil, gas, uranium) formed over millions of years and are consumed far faster than they re-form — so the stock shrinks. Renewables ride an ongoing flow (sunlight, wind, the water cycle, the tides, Earth's internal heat, this season's plant growth) that keeps arriving.

◆ NOTE

Analogy · stock vs flow Non-renewables are a savings account you can only withdraw from — every kWh spent is gone. Renewables are a salary that lands every month: spend within the inflow and you never go broke. Coal is the inheritance; sunshine is the paycheck.

◆ NOTE

Trap · renewable ≠ zero-emission A source can be renewable but still emit greenhouse gases (burning biomass releases CO₂), and a renewable can carry big environmental costs (a hydro dam floods habitat).

"Renewable" describes replenishment rate, not cleanliness. Keep the two ideas separate in exam answers.

The seven sources, one card each

For every source, learn the same four things: the natural flow it taps, the energy-conversion chain to electricity, why it's renewable, and its standout advantage / disadvantage. The conversion chains matter — they connect straight to KK05.

◆ NOTE

Memory hook "Sun-Wind-Water-Tide, Heat-and-Harvest" — solar & wind (intermittent), hydro & tidal (water-driven), geothermal (Earth's heat), then the two harvest sources: biomass (solid) and biofuels (liquid). Seven sources, three families.

:00 · CLEAR & BREEZY * CLEAR/BREEZY ▲ CLOUDY/STILL 🌪 WINDY STORM 3D NOT AVAILABLE — USE THE READOUT PANEL BELOW; IT SHOWS THE SAME LIVE OUTPUT VALUES FOR EACH SOURCE.

06 3D · The intermittency lab

Renewables don't all behave the same across a day. Drag the time-of-day slider and flip the weather — watch which sources surge, fade or stay rock-steady. This is the single most examined idea in the chapter, and it sets up base/peak load (KK09).

SOLID · BURNED DIRECTLY

Biomass vs biofuels — the classic mix-up

The study design lists these as two separate sources, and examiners love testing whether you can tell them apart. Both come from recently living matter; the difference is form and processing.

Organic matter burned to release stored chemical energy as heat. Wood, crop stubble, bagasse (sugarcane fibre), forestry residue, even municipal organic waste. Burned to raise steam → turbine → generator.

Mirran Plains example: a small bagasse/stubble boiler at the edge of the cane and wheat district burns harvest residue that would otherwise rot.

Liquid or gaseous fuels processed from biomass to power engines — ethanol (from sugar/starch, blended as E10) and biodiesel (from canola or used cooking oil). Biogas (methane) from anaerobic digestion also counts.

Mirran Plains example: local canola → biodiesel runs the farm machinery and a backup generator; the servo sells E10.

◆ **NOTE**

The "carbon-neutral in principle" idea Plants absorb CO_2 as they grow, then release roughly that same CO_2 when burned — so the combustion carbon can be close to balanced if new crops regrow to reabsorb it. That's why biomass/biofuels are called approximately carbon-neutral. The catch examiners reward: it's only near-neutral once you subtract the fossil energy used to grow, harvest, process and transport the feedstock — and not if forests are cleared faster than they regrow.

Intermittent vs dispatchable — the deciding axis

Beyond "renewable or not", the property that drives real planning decisions is whether a source can deliver power on demand. This single table answers most comparison questions in the chapter.

Source	Natural flow	Renewable?	On-demand (dispatchable)?	Greenhouse at point of use
Solar PV	Sunlight	Yes	No — intermittent (day/cloud)	~Zero
Wind	Moving air	Yes	No — intermittent (wind speed)	~Zero
Hydro-electric	Falling/stored water	Yes	Yes — fast, storable	~Zero*
Tidal	Ocean tides	Yes	Predictable but cyclic	~Zero
Geothermal	Earth's internal heat	Yes	Yes — steady baseload	Low
Biomass	This season's plant growth	Yes	Yes — fuel can be stored	CO_2 released ($\approx$ reabsorbed)
Biofuels	Crops → liquid fuel	Yes	Yes — fuel stored	CO_2 released ($\approx$ reabsorbed)

*Reservoir hydro can emit methane from flooded vegetation, especially in warm climates — a nuance worth a mark if a question pushes on it.

◆ **NOTE**

Why this matters Solar and wind are cheap and clean but intermittent — they generate when nature decides, not when we flick the switch. Hydro (especially pumped hydro like Snowy 2.0), geothermal, biomass and stored biofuels are dispatchable — they can fill the gaps. A good energy mix pairs cheap intermittent generation with dispatchable backup and storage. Hold this thought for KK09.

Command-word decoder

◆ COMMAND WORDS

Match what the verb is asking for. Tap a word to expand.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

🔗 P·E·L WRITING

The house writing frame: Point → Explain → Link. Draft a response to the prompt below; the trainer lights up as you hit the ideas examiners reward. Prompt: Explain why solar and wind are described as intermittent, and why this makes a dispatchable renewable like hydro-electricity valuable in the same grid. (4 marks) State that solar/wind depend on a natural condition that varies, so output is not constant. Link the variation to the source (no sun at night/cloud; no output when wind is too low/high) and contrast with water that can be stored and released on demand. Conclude that dispatchable hydro fills the gaps, keeping supply matched to demand — directly answering "why valuable".

Good vs bad answer, side by side

Question: Compare a solar farm and a geothermal plant as sources of electricity for the Mirran Plains zone, referring to renewability and reliability of supply. (4 marks)

• NOTE

Vague, one-sided, no real comparison. "Both renewable" isn't explained, supply reliability is ignored, and the geothermal point is dropped. "Compare" needs both sources weighed against the same criteria.

• NOTE

Both sources weighed on the same two criteria, the intermittent/dispatchable contrast is explicit, and it lands a judgement tied to the scenario. Full marks.

Practice exam questions

★ PRACTICE & SELF-MARK

VCAA-style questions totalling 20 marks. Write your answer, reveal the model and marking guide, then self-mark with the stars. Your progress saves automatically.

One-screen cheat sheet

♦ RECAP / CHEAT SHEET

Replenished by natural processes at a rate $\geq$ the rate of use, so not depleted on a human timescale. About rate, not cleanliness. Solar, wind, hydro-electric, tidal, geothermal, biomass, biofuels. Three families: sun/wind, water (hydro/tidal), then Earth-heat + harvest. Solar & wind — generate when nature allows, not on demand. Output varies with sun/cloud and wind speed. Hydro, geothermal, biomass, stored biofuels — power on demand. Tidal is predictable but cyclic. Biomass = solid matter burned directly. Biofuel = liquid/gas fuel refined from biomass (ethanol, biodiesel, biogas). Bio sources release CO_2 but regrowth reabsorbs $\approx$ the same — near-neutral, minus fossil energy used to grow/process/transport. Wind/hydro/tidal: kinetic $\rightarrow$ electrical. Solar PV: radiant $\rightarrow$ electrical (no turbine). Geo/biomass/biofuel: thermal $\rightarrow$ kinetic $\rightarrow$ electrical. Snowy 2.0 pumped hydro (2.2 GW, dispatchable storage); Golden Plains wind (VIC); large solar farms (Western Downs, New England); geothermal direct-use; ethanol E10 & canola biodiesel.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer shape — make a Point, support with Evidence, then Link to the question.

X WEAK ANSWER

Solar is good because it's clean and free from the sun. Geothermal uses heat from the ground which is also good. Both are renewable so they're better than coal. The Mirran Plains zone should use solar because Australia is sunny. Vague, one-sided, no real comparison. "Both renewable" isn't explained, supply reliability is ignored, and the geothermal point is dropped. "Compare" needs both sources weighed against the same criteria.

✓ STRONG / MODEL ANSWER

Both are renewable: sunlight and Earth's internal heat are replenished far faster than they are used. They differ on reliability of supply. Solar PV is intermittent — it generates only in daylight and drops under cloud, so its output varies and cannot be guaranteed at peak evening demand. Geothermal taps a continuous heat flow, so it runs steadily day and night as dispatchable baseload. For Mirran Plains, solar offers cheap bulk daytime energy, while geothermal provides the reliable around-the-clock supply solar lacks — so the two are complementary rather than competing. Both sources weighed on the same two criteria, the intermittent/dispatchable contrast is explicit,